

Exercise Sheet 10b

Predicate Logic – Natural Deduction & Semantics

Consider the following signature:

- Function symbols: **zero** (arity 0); **succ** (arity 1)
- Predicate symbols: $<$ (arity 2); $\leq$ (arity 2)

We will use infix notation for the binary symbols $<$ and $\leq$. For simplicity we write 0 for **zero**, 1 for **succ(zero)**, 2 for **succ(succ(zero))**, etc. Consider the following formulas that capture properties of the above symbols:

- let S_1 be $\forall x. \exists y. x < y$
- let S_2 be $\forall x. \forall y. x < y \rightarrow \text{succ}(x) \leq y$
- let S_3 be $\exists x. 1 \leq x$

1. Provide a constructive Natural Deduction proof of $(S_1) \rightarrow \neg \exists x. \forall y. \neg x < y$
2. Provide a Constructive Natural Deduction proof of $(S_1) \rightarrow (S_2) \rightarrow S_3$
3. Provide a model M_1 such that $\models_{M_1} S_1$
4. Provide a model M_2 such that $\models_{M_2} \neg S_1$